

## Assignment 2

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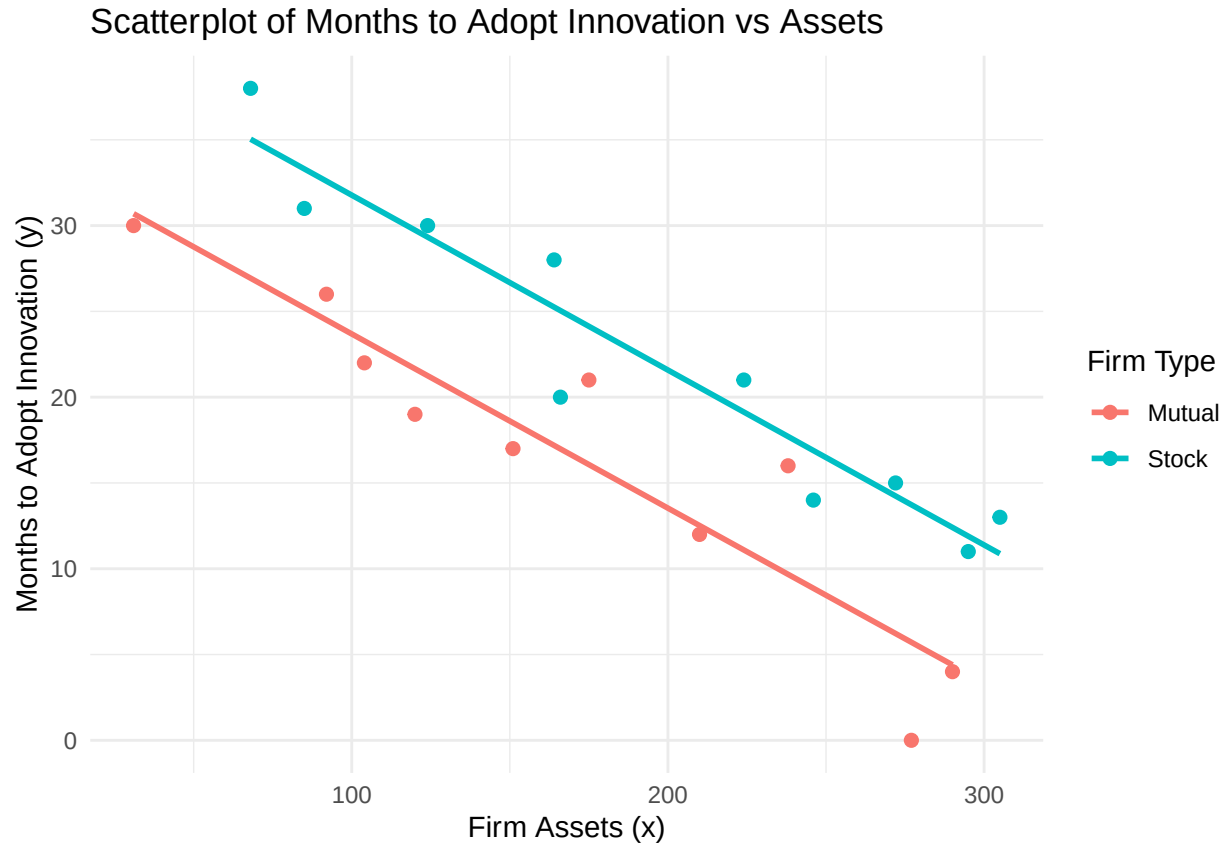
2025-07-20

**Question 1:** Using the Insurance Innovation Data in InsInnov.xlsl, y is the number of months it took to adopt a specific innovation in a firm. The predictor variables being considered is x, the assets of the firm, and the type of the firm.

(a) Construct a scatter diagram of y vs x and add a line of best fit (using ggplot) for both types of firms.

```
# Install and load packages
# install.packages("ggplot2")
library(ggplot2)
library(readxl)
# Load the data
data <- read_excel("InsInnov.xlsx")
# Create the scatter plot
ggplot(data, aes(x = X, y = Y, color = Type)) +
  geom_point(size = 2) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Scatterplot of Months to Adopt Innovation vs Assets",
    x = "Firm Assets (x)",
    y = "Months to Adopt Innovation (y)",
    color = "Firm Type"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



(b) comment briefly on whether you think each model will be a good fit to the data, and explain your reasoning.

- Model 1: This model assumes a single line fits all the data, ignoring firm type. But the plot shows two different lines for Mutual and Stock firms with different intercepts and slopes. So this model would not fit well because it oversimplifies things and ignores group differences.
- Model 2: This adds a dummy variable for firm type, so now it allows two lines with different intercepts but same slope. That's a better fit than Model 1 because it captures the difference between Stock and Mutual firms however if you look closely at the scatterplot, the slopes aren't exactly parallel. So this model still isn't quite right.
- Model 3: This model is the best fit. It includes an interaction term ( $xD$ ), which allows different slopes and intercepts for each firm type. Since the scatterplot clearly shows the lines going in different directions, this model reflects the pattern we actually see in the data.

(c) Run a regression summary for Model 2.

```
# Create dummy variable for Type
data$D <- ifelse(data$Type == "Stock", 1, 0)

# Fit Model 2
model2 <- lm(Y ~ X + D, data = data)
```

```
# Show summary
summary(model2)
```

```
##
## Call:
## lm(formula = Y ~ X + D, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.6915 -1.7036 -0.4385  1.9210  6.3406
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 33.874069   1.813858  18.675 9.15e-13 ***
## X           -0.101742   0.008891 -11.443 2.07e-09 ***
## D             8.055469   1.459106   5.521 3.74e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.221 on 17 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8827
## F-statistic: 72.5 on 2 and 17 DF,  p-value: 4.765e-09
```

- $H_0: B_2 = 0$  (firm type has no effect on adoption time)
- $H_a: B_2 \neq 0$  (firm type does have an effect)
- Significance level:  $\alpha = 0.05$
- $p\text{-value} = 0.0000374$
- We're testing if firm type actually affects how long it takes to adopt innovation. The  $p\text{-value}$  for the dummy variable is  $3.74e-05$ , which is way smaller than  $0.05$ , so we reject the null hypothesis.
- That means firm type does matter because stock firms take about 8 months longer ( $8.055469$ ) to adopt innovation than mutual firms, even when they have the same level of assets.

(d) For model 2, compute a 95% confidence interval for  $B_2$  and interpret the results.

```
# Get the 95% confidence interval for B2
confint(model2, level = 0.95)
```

```
##              2.5 %      97.5 %
## (Intercept) 30.0471625 37.70097553
## X           -0.1205009 -0.08298329
## D             4.9770253 11.13391314
```

- The 95% confidence interval for  $B_2$  is from about 4.98 to 11.13. So basically, we're 95% confident that stock firms take somewhere between 5 and 11 more months to adopt innovation compared to mutual firms, after accounting for their assets.
- Since the whole interval is above zero, that means firm type actually does have a real impact since stock firms really do take longer.

(e) Run a regression summary for Model 3. With the addition of the interaction term, what does the p-value related to that term imply for the model. Is the interaction term necessary in Model 3? Is this consistent with the scatter diagram in (a).

```
# Fit Model 3 (with interaction term)
model3 <- lm(Y ~ X + D + X:D, data = data)

# Show summary
summary(model3)

##
## Call:
## lm(formula = Y ~ X + D + X:D, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.7144 -1.7064 -0.4557  1.9311  6.3259
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 33.8383695  2.4406498  13.864 2.47e-10 ***
## X           -0.1015306  0.0130525  -7.779 7.97e-07 ***
## D             8.1312501  3.6540517   2.225  0.0408 *
## X:D          -0.0004171  0.0183312  -0.023  0.9821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.32 on 16 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8754
## F-statistic: 45.49 on 3 and 16 DF,  p-value: 4.675e-08
```

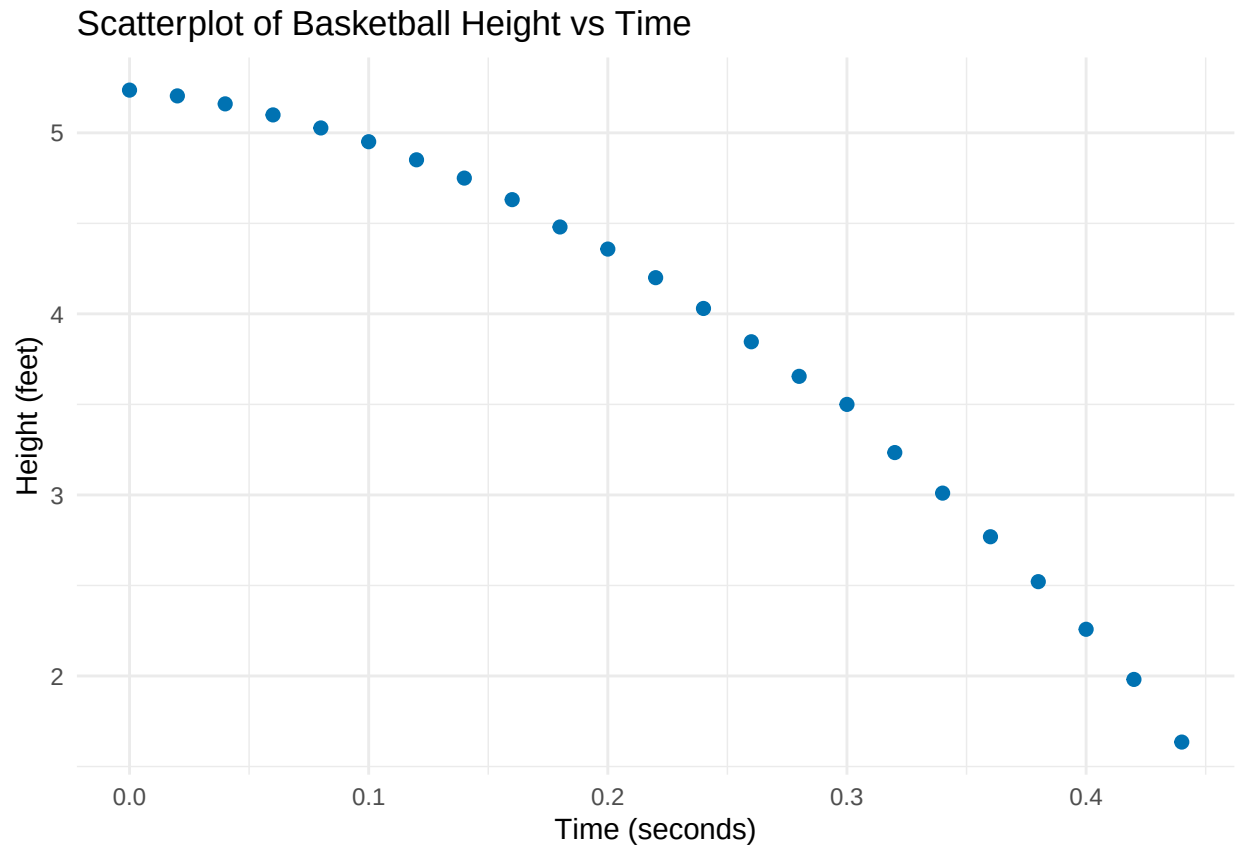
- So after running the regression for Model 3, the interaction term (X:D) has a p-value of 0.9821, which is super high so definitely not significant. That means the interaction term doesn't actually help explain anything in the model. Basically, the slopes for mutual and stock firms are the same, and the only difference is their intercepts.

**Question 2:** A basketball is dropped from a height of about 5.25 feet. The height of the basketball is recorded 23 times at intervals of about 0.02 seconds. The results are in the Basketball.xlsx file.

(a) Construct a scatter diagram of Height (y) vs Time (x)

```
# Read the Excel file
data <- read_excel("Basketball.xlsx")
# Scatterplot of Height vs Time
ggplot(data, aes(x = Time, y = Height)) +
  geom_point(color = "#0072B2", size = 2) +
```

```
labs(title = "Scatterplot of Basketball Height vs Time",
     x = "Time (seconds)",
     y = "Height (feet)") +
theme_minimal()
```



(b) Which time of model would be the best fit for this data? Explain.

- The points clearly curve downward meaning it's not a straight line, so a linear model wouldn't fit. It looks like a parabola, so a quadratic model makes the most sense.

(c) Run a regression summary on the model chosen in (b).

```
# Create a new variable for Time squared
data$Time2 <- data$Time^2

# Fit the quadratic regression model
model_quad <- lm(Height ~ Time + Time2, data = data)

# View the regression summary
summary(model_quad)
```

```
##
```

```
## Call:
## lm(formula = Height ~ Time + Time2, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.038774 -0.004880 -0.000209  0.004238  0.043420
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.23125     0.00871   600.62 < 2e-16 ***
## Time         -1.26642     0.09170  -13.81 1.09e-11 ***
## Time2        -15.49718     0.20127  -77.00 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01515 on 20 degrees of freedom
## Multiple R-squared:  0.9998, Adjusted R-squared:  0.9998
## F-statistic: 6.06e+04 on 2 and 20 DF,  p-value: < 2.2e-16
```

(d) Using the p-values of the coefficient, would you consider each of these models components are significant? Explain.

- All the p-values are way smaller than  $\alpha = 0.05$ , which means each term is statistically significant. The intercept, time, and time<sup>2</sup> all actually matter and help predict the basketball's height.

(e) Predict the time when the basketball will hit the ground.

```
# Fit quadratic regression model
model <- lm(Height ~ Time + I(Time^2), data = data)
# Get coefficients
coefs <- coef(model)
a <- coefs[3] # Coefficient for Time^2
b <- coefs[2] # Coefficient for Time
c <- coefs[1] # Intercept

# Solve for Time when Height = 0 using quadratic formula
root1 <- (-b + sqrt(b^2 - 4*a*c)) / (2*a)
root2 <- (-b - sqrt(b^2 - 4*a*c)) / (2*a)

# Choose the meaningful root (positive time)
root1
```

```
##      Time
## -0.6232952
```

```
root2
```

```
##      Time
## 0.5415758
```

- The basketball hits the ground at around 0.54 seconds after it was dropped.

**Question 3:** The United States Bureau of Labour Statistics produces consumer price indexes (CPIs) for several industries. Of interest is to develop a model using backward elimination with Alpha stay = 0.10 to predict the percentage change in CPI (from the previous years) for all commodities,  $y$ , as a function of the percentage change in the CPI for a number of candidate industries: Food ( $x_1$ ), Shelter ( $x_2$ ), Apparel ( $x_3$ ), and Fuel Oil ( $x_4$ ). Data on the annual percentage change in CPI for all commodities and the candidate industries for 20 randomly selected years is found in Commodities.xlsx.

(a) Build a linear regression model including all candidate industries. Output the summary table of the model.

```
# Read in the Commodities dataset
data <- read_excel("Commodities.xlsx")
# View column names to confirm
colnames(data)
```

```
## [1] "Y=AllCommodities" "X1=Food"          "X2=Shelter"      "X3=Apparel"
## [5] "X4=FuelOil"
```

```
# Fit the full model
model_full <- lm(`Y=AllCommodities` ~ `X1=Food` + `X2=Shelter` + `X3=Apparel` + `X4=FuelOil`, data = data)
# Display summary output
summary(model_full)
```

```
##
## Call:
## lm(formula = `Y=AllCommodities` ~ `X1=Food` + `X2=Shelter` +
##     `X3=Apparel` + `X4=FuelOil`, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.24737 -0.39537  0.02015  0.27118  1.29783
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.059031   0.389911  -2.716   0.0159 *
## `X1=Food`      0.475326   0.062879   7.559 1.72e-06 ***
## `X2=Shelter`   0.249518   0.100547   2.482   0.0254 *
## `X3=Apparel`  -0.008375   0.162620  -0.052   0.9596
## `X4=FuelOil`   0.272111   0.115423   2.358   0.0324 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7472 on 15 degrees of freedom
## Multiple R-squared:  0.9631, Adjusted R-squared:  0.9533
## F-statistic: 97.98 on 4 and 15 DF, p-value: 1.461e-10
```

(b) Applying backward elimination, is there any independent variables in the model that you would remove, or should all stay? If your answer is “yes”, which variables would you remove? Why? (do not actually run the backward elimination process, just state what you would do).

- Looking at the p-values in the full model summary, I would apply backward elimination by removing any variables that aren’t significant at the 0.10 level.
- I would remove Apparel from the model because its p-value is way above 0.10, meaning it’s not statistically significant. The other variables like Food, Shelter, and Fuel Oil, all have p-values less than 0.10, so they should stay.

(c) Given your answer in (b), what would the next step be? (do not actually run the backward elimination process, just state what you would do).

The next step would be to remove Apparel from the model and rerun the regression with just Food, Shelter, and Fuel Oil. Then I’d check the new p-values to see if any other variables need to be removed.

(d) Perform backward elimination to your full model. What model did you end with?

```
# Rename columns for modeling
colnames(data) <- c("AllCommodities", "Food", "Shelter", "Apparel", "FuelOil")

# Fit full model with renamed columns
model_full <- lm(AllCommodities ~ Food + Shelter + Apparel + FuelOil, data = data)

# Perform backward elimination
ols_step_backward_p(model_full , p_val= 0.10)
```

```
##
##
##                               Stepwise Summary
## -----
## Step    Variable      AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0       Full Model    51.346   57.320   -2.300   0.96314   0.95331
## 1       Apparel      49.350   54.328   -4.965   0.96313   0.95622
## -----
##
## Final Model Output
## -----
##
##                               Model Summary
## -----
## R                0.981      RMSE                0.647
## R-Squared        0.963      MSE                  0.419
## Adj. R-Squared   0.956      Coef. Var          16.807
## Pred R-Squared   0.917      AIC              49.350
## MAE              0.487      SBC              54.328
```

```
## -----
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
##
## ANOVA
## -----
## Sum of
## Squares      DF      Mean Square      F      Sig.
## -----
## Regression    218.794      3      72.931    139.32    0.0000
## Residual       8.376     16       0.523
## Total        227.170     19
## -----
##
## Parameter Estimates
## -----
## model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
## -----
## (Intercept) -1.062      0.375           -2.834      0.012      -1.856      -0.267
## Food         0.474      0.057           0.596      8.320      0.000      0.353      0.595
## Shelter      0.249      0.097           0.267      2.562      0.021      0.043      0.456
## FuelOil      0.269      0.096           0.234      2.808      0.013      0.066      0.472
## -----
```

- I ran backward elimination with Alpha = 0.10 and ended up dropping Apparel from the model because it wasn't statistically significant. The final model includes Food, Shelter, and FuelOil since they all had p-values below 0.10.

(e) Perform stepwise regression with Alpha (enter) = 0.10 Alpha (stay) = 0.10. Describe the sequence of tests that were performed.

```
# Load the olsrr package
library(olsrr)
# Rename columns for modeling
colnames(data) <- c("AllCommodities", "Food", "Shelter", "Apparel", "FuelOil")
# Fit the full model
model_full <- lm(AllCommodities ~ Food + Shelter + Apparel + FuelOil, data = data)
# Perform stepwise regression (matching Lesson 4 style)
ols_step_both_p(model_full, pent = 0.10, prem = 0.10)
```

```
##
##
## Stepwise Summary
## -----
## Step      Variable      AIC      SBC      SBIC      R2      Adj. R2
## -----
## 0      Base Model      109.357    111.348    48.889    0.00000    0.00000
## 1      Food (+)      74.965     77.952    14.449    0.83791    0.82890
```

```
## 2      FuelOil (+)      54.226      58.209      -2.866      0.94800      0.94188
## 3      Shelter (+)      49.350      54.328      -4.965      0.96313      0.95622
## -----
##
## Final Model Output
## -----
##
##                               Model Summary
## -----
## R                               0.981      RMSE                               0.647
## R-Squared                       0.963      MSE                               0.419
## Adj. R-Squared                  0.956      Coef. Var                       16.807
## Pred R-Squared                  0.917      AIC                               49.350
## MAE                             0.487      SBC                               54.328
## -----
## RMSE: Root Mean Square Error
## MSE: Mean Square Error
## MAE: Mean Absolute Error
## AIC: Akaike Information Criteria
## SBC: Schwarz Bayesian Criteria
##
##                               ANOVA
## -----
##                               Sum of
##                               Squares      DF      Mean Square      F      Sig.
## -----
## Regression      218.794      3      72.931      139.32      0.0000
## Residual         8.376      16      0.523
## Total           227.170      19
## -----
##
##                               Parameter Estimates
## -----
## model      Beta      Std. Error      Std. Beta      t      Sig      lower      upper
## -----
## (Intercept) -1.062      0.375      -2.834      0.012      -1.856      -0.267
## Food         0.474      0.057      0.596      8.320      0.000      0.353      0.595
## FuelOil      0.269      0.096      0.234      2.808      0.013      0.066      0.472
## Shelter      0.249      0.097      0.267      2.562      0.021      0.043      0.456
## -----
```

- To perform stepwise regression with entry and stay criteria of  $\alpha = 0.10$ , I used the `ols_step_both_p()` function. The procedure started with the base model (no predictors) and evaluated which variables to add based on their significance. Food was added to the model first because it had the strongest significance (lowest p-value). FuelOil was added next, as it met the entry threshold ( $p < 0.10$ ). Shelter was added last, also meeting the  $\alpha = 0.10$  criteria. Apparel was not added at any step, as it did not meet the entry requirement.
- The final model included Food, FuelOil, and Shelter, with an Adjusted  $R^2$  of 0.956 and all predictors statistically significant at the 0.05 level.

(f) What independent variables are in the final model suggested by stepwise regression? Is this the same model as using backward elimination?

- The final model from stepwise regression includes the following independent variables: Food, FuelOil, Shelter
- Yes, this is the same model that resulted from the backward elimination process. In both cases, Apparel was excluded because it was not statistically significant at the 0.10 level.

(g) Which independent variables appear to be the most important in the final model? Explain.

- The most important variable in the final model is Food because it has the highest standardized beta coefficient (0.596), which means it has the biggest impact on the dependent variable.

(h) Perform the best subset regression. Which model would you recommend from the best subset regression procedure? Explain.

```
# Best subset regression
ols_step_best_subset(model_full)
```

```
##           Best Subsets Regression
## -----
## Model Index   Predictors
## -----
##      1         Food
##      2      Food FuelOil
##      3      Food Shelter FuelOil
##      4      Food Shelter Apparel FuelOil
## -----
##
##                                     Subsets Regression Summary
## -----
## Model      R-Square    Adj.      Pred      C(p)      AIC      SBIC      SBC      MSEP
## -----
##      1         0.8379      0.8289      0.7511    49.9561    74.9648    14.4493    77.9520    40.9375
##      2         0.9480      0.9419      0.9046     7.1585    54.2259     -2.8661    58.2088    13.9534
##      3         0.9631      0.9562      0.9169     3.0027    49.3496     -4.9651    54.3283    10.5534
##      4         0.9631      0.9533      0.8995     5.0000    51.3461     -2.3004    57.3205    11.3052
## -----
## AIC: Akaike Information Criteria
## SBIC: Sawa's Bayesian Information Criteria
## SBC: Schwarz Bayesian Criteria
## MSEP: Estimated error of prediction, assuming multivariate normality
## FPE: Final Prediction Error
## HSP: Hocking's Sp
## APC: Amemiya Prediction Criteria
```

- I'd go with Model 3 because it includes Food, Shelter, and FuelOil. It has the highest adjusted R-squared (0.9562), the lowest AIC (49.35), and the lowest C(p) value (3.0027). Adding Apparel in Model 4 doesn't help, so it's better to leave it out and stick with the simpler, stronger model.

(i) If you had to pick a model with only two independent variables what would you choose? Explain

- I'd pick model 2 with Food and FuelOil. It has a high adjusted R-squared of 0.9419, low AIC and SBIC values, and a much better predictive power (Pred R-Square = 0.9046) compared to the other two-variable option. It gives strong results without overcomplicating the model.

(j) Suppose that you decided on a final model containing Food, Shelter, and Fuel Oil. Is there any serious multicollinearity present? Explain

```
# Fit the model with Food, Shelter, and Fuel Oil
model_vif <- lm(AllCommodities ~ Food + Shelter + FuelOil, data = data)
# Calculate and display VIF and tolerance
ols_vif_tol(model_vif)
```

```
##   Variables Tolerance      VIF
## 1      Food 0.4485763 2.229275
## 2    Shelter 0.2126820 4.701855
## 3    FuelOil 0.3323207 3.009141
```

- No, there isn't any serious multicollinearity. All VIF values are below 5, with the highest being 4.7 for Shelter.

(k) Using the model with Food, Apparel, and Fuel Oil

(i) Are there any observations that should be classified as high leverage? Explain.

(ii) Are there any outliers? Explain.

(iii) Are any of the observations listed in (i) or (ii) influential? Explain.

```
# Fit the model with Food, Apparel, and Fuel Oil
model_k <- lm(AllCommodities ~ Food + Apparel + FuelOil, data = data)

# (i) High leverage: Calculate hat values and plot
leverage <- hatvalues(model_k)
cutoff_leverage <- 2 * (length(coefficients(model_k)) / nrow(data))
which(leverage > cutoff_leverage)
```

```
## 14 15
## 14 15
```

- Observations 14 and 15 have high leverage because their leverage values are above the cutoff. This means they have a strong influence on the fitted values.

```
# (ii) Outliers: Standardized residuals
standardized_resid <- rstandard(model_k)
which(abs(standardized_resid) > 2)
```

```
## 14 20
## 14 20
```

- Observations 14 and 20 are outliers since their standardized residuals are greater than 2 in absolute value. This suggests their actual values differ a lot from what the model predicted.

```
# (iii) Influential points: Cook's distance
cooks <- cooks.distance(model_k)
which(cooks > 1)
```

```
## 14
## 14
```

- Observation 14 is influential based on Cook's Distance being greater than 1. This means it affects the regression model significantly and might be worth investigating.

## Question 4: Comparing Model Selection Methods

**(a) Explain the theoretical underpinnings of the forward selection, backward elimination, and stepwise selection methods. Discuss how these methods balance the trade-off between model complexity and goodness-of-fit.**

- Forward selection starts with no variables and adds them one by one. It picks the variable that improves the model the most and keeps going until no more are useful. It helps keep the model simple.
- Backward elimination starts with all the variables and removes the least useful ones one at a time. It keeps going until only the significant variables are left. This helps get rid of things that don't really help.
- Stepwise is a mix of both. It adds and removes variables along the way. If something added earlier turns out not to help, it can take it out later.
- All three methods try to make a model that fits the data well without making it too complicated. They help find a good balance between accuracy and simplicity.

**(b) Consider a scenario where you have a dataset with a large number of predictors. Discuss the potential pitfalls of using forward selection and backward elimination in this context. What are the advantages and disadvantages of using the stepwise selection method instead?**

- If you have a dataset with a lot of predictors, forward selection might miss important variables because it only adds one at a time based on what looks best right away. It doesn't look at the bigger picture or how variables work together.
- Backward elimination can be a problem too because starting with all the predictors can make the model too big at first. If there's multicollinearity or too many weak variables, it can be hard to know what to remove first.
- Stepwise tries to fix that by letting you add and remove variables throughout the process. That can lead to a better model, but it still doesn't test every combo and might miss the absolute best one. Also, if there's too much back and forth, it can overfit the data.

## Question 5: Partial F Test

(a) Define the partial F test and explain its role in comparing nested models. How does it help in determining the significance of adding or removing predictors from a model?

- The partial F test is used when you want to compare two models, one with more variables (full model) and one with fewer (reduced model). It tells you if the extra variables in the full model actually make a big enough difference. If they do, you keep them. If not, you can drop them and go with the simpler model. It's a way to test if adding or removing predictors really matters.

(b) Calculate the F-statistic for the partial F test and interpret the results. Discuss the implications of your findings on model selection and the potential impact of multicollinearity and model stability.

```
# Given values
SS_A <- 500
SS_B <- 520
k_A <- 5
k_B <- 3
n <- 100

# Calculate F-statistic
numerator <- (SS_B - SS_A) / (k_A - k_B)
denominator <- SS_A / (n - k_A - 1)
F_stat <- numerator / denominator

# Print result
F_stat
```

```
## [1] 1.88
```

- Based on the partial F test, the F-stat is 1.88, which is not that high. This means that adding X4 and X5 does not significantly improve the model. So I wouldn't keep them.
- This suggests that the simpler model with just X1, X2, and X3 is good enough, and adding extra predictors might just make things more complicated without real benefit.
- When there's a lot of predictors, sometimes they overlap in the info they give, which can lead to multicollinearity. That makes it hard to trust your coefficients and can mess with how stable your model is.